# STUDENT PERFORMANCE ANALYSIS SYSTEM

PROJECT REPORT

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This report describes the features present in the supplied website and clearly identifies proposed extensions that are not yet implemented.

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# A new student performance analysing system using knowledge discovery in higher educational databases

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ABSTRACT

Knowledge discovery is a wide ranged process including data mining, which is used to find out meaningful and useful patterns in large amounts of data. In order to explore the factors having impact on the success of university students, knowledge discovery software, called MUSKUP, has been developed and tested on student data. In this system a decision tree classification is employed as a data mining technique. With this software system all the tasks involved in the knowledge discovery process are kept together. The advantage of this approach is to have access to all the functionalities of SQL server and Analysis Services through single software. The study was carried out on the data from university students. According to results of the study, the types of registration to the university and the income levels of the students' family were found to be associated with student success.

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## 1. Introduction

Now, amount of data is doubling every 3 years (Lyman & Hal, 2003). As the amount of data increases, it is necessary to analyze it faster and obtain meaningful and operational information. In the analysis of large amounts of data, former generation statistical analysis techniques are viewed to be inadequate (Glymour et al., 1996). Therefore, they are replaced by new generation techniques which are called knowledge discovery in databases (KDD) and data mining (DM) (Munakata, 1999). These new techniques aid the human being intelligently to discover valuable information in data analysis.

KDD is a process which is used to discover patterns in data (Agrawal et al., 1993; Brachman et al., 1996; Fayyad & Uthurusamy, 1996; Mitchell, 1999). Here, pattern means potentially useful and interesting tendencies and relationships in data. DM is extracting patterns from data. It is commonly used in a wide range of profiling practices, such as marketing, surveillance, fraud detection and scientific discovery (Han & Kamber, 2006). Recently the discovery methodologies were used to enhance and evaluate the higher education tasks.

Besides having the role of producing and disseminating information, universities are fundamentally institutions that educate students. Universities record data containing valuable information about students, which are usually only used individually and for official uses (transcript, registration, etc.). However, these data could be used in academic consulting of students. In order to do so, they need to be analyzed, which requires a separate discovery investigation (Guvenc, 2001).

In this direction, some models have been proposed and implemented. Delavari and Beikzadeh (2004) have proposed a model to represent how DM can be used in a higher educational system to improve the efficiency and effectiveness of the traditional processes. In the model, several processes are proposed to be enhanced through DM functions. The model is also presented as a guideline for higher educational system to improve the decision making processes. Potawat et al. (2003) have used rough set theory as a classification approach to analyze student data to describe different dependencies between the attributes and the student status. The model of Delavari et al. (2005) is used as a roadmap for the application of DM in higher educational system.

The research by Mierle et al. (2005) has very specifically approach that describes the results of analysing data from a large collection of the so called concurrent version system created by many students working on a small set of identical projects (course assignments) in the second year undergraduate computer science course. The proposed model is used to extract all information of student behavior in writing

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the code of assignments and to find some statistical patterns or predicators that can be used to enhance students' performance in writing the code.

In this study (Guruler, 2005) knowledge discovery was conducted on the available databases to evaluate the current academic standings of the students and to make viable predictions for the future. As a part of this study, MUSKUP (Mugla University Student Knowledge discovery Unit Program), a system that integrates the knowledge discovery process with the Database Management System (DBMS), was developed. MUSKUP is a basic-level knowledge discovery tool through which information is obtained by using students' demographical data.

This paper introduces using the KDD process of which attributes may affect the student success in education.

### 1.1. Knowledge discovery process

The steps involved in KDD process are shown in Fig. 1 (Fayyad & Uthurusamy, 1996). In the *data selection* step, a sample data set appropriate for the selected problem is obtained by combining a several data sets. Then, in the *pre-processing* step the irrelevant elements in the selected sample, erroneous records, and null columns (variables) are eliminated, the repetitive records are singled, and it is decided on whether or not the transformations on some columns are necessary, etc. This step improves the quality of the discovered information. After the pre-processing step, the discrepancies in the selected sample are eliminated and the new variables are obtained from available variables in the *data transformation* step. This step helps to enhance the validity of the model formed in DM. In DM, a selected DM algorithm such as classification, grouping, and matching is applied on the pre-processed data. Maimon and Rokach (2005) offers a survey about Algorithms for DM and KDD. Finally, the discovered information is evaluated with regards to its validity, usefulness, and simplicity in the *validation* step.

## 2. Methodology

This study aims to discover individual student characteristics that are associated with their success by using a Microsoft Decision Trees (MDT) classification technique. Here, each student is categorized as either successful or unsuccessful according to grade point averages (GPA). Here are the steps to study.

### 2.1. Preparation of database

First of all, database should be prepared and designed for knowledge discovery (Chen et al., 1996). In this study knowledge discovery research was conducted on demographical data of the students attending to the Faculty of Economics and Social Sciences of Mugla University from 1995 on. The fact that the departments of the faculty were highly similar to each other, that the faculty was the oldest and the most sophisticated faculty of the university and that it had most students caused this faculty to be chosen to be included in this study.

Student data obtained from the database of the directory of student affairs of the university consist of many tables. Thirteen tables which were related to the scope of this study were selected. We related these tables with each other and filtered them in such a way that a single *view* on the same database was formed. In this *view* there are a total of 111 columns.

The data used in the discovery is mostly obtained while students have been registered to the university. These are: the registered information for state (registered city, date of birth, etc.), high school information (diploma degree, education type, foreign language, etc.), Turkish university entrance exam degree and university placement information, family's living conditions and financial status. Except that students' department information (such as the number of the terms spent at school) at the university and GPA values.

Since GPAs of the students are the best indicators for the success level of the education, the column containing GPAs was used as target column in establishing the model. All of the columns in the data set were used to find out which column or columns best described the patterns of successful/unsuccessful students.

### 2.2. MUSKUP (Mugla University Student Knowledge discovery Unit Program)

In the solution development phase of the study, a task sharing mechanism between SQL server and Analysis Services has been developed so as to implement the tasks involved in every individual KDD step. In fact, each of the tasks can separately be implemented either on SQL server or Analysis Services. In order to perform all the tasks together MUSKUP has been developed. The purposes of MUSKUP are:

- Setting an example for KDD process.
- Having access to all the tasks from single program.

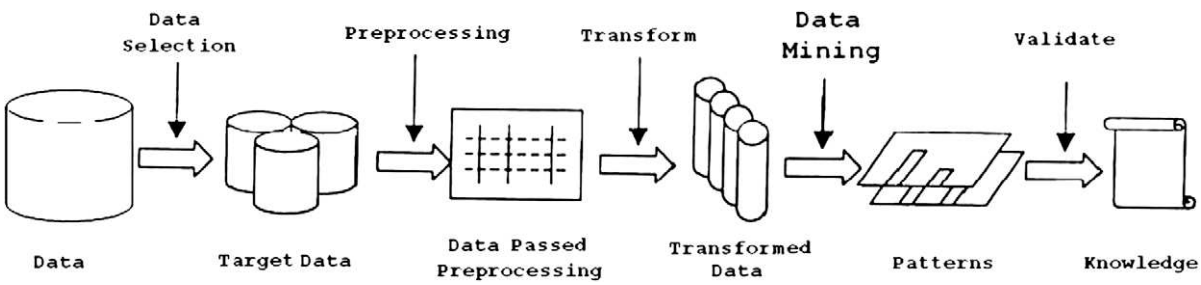



**Fig. 1.** Knowledge discovery process.

- Having access to functions of SQL server and Analysis Services by means of programming techniques. In this way the user can perform tasks without any need for complicated SQL statements.

There are many commercial software applications for DM or KDD (Goebal & Grunwald, 1999; (The Data and Analysis Center for Software) DACS, 2009). MUSKUP has been developed specifically for student demographical data.

The MUSKUP user interface is shown in Fig. 2. The steps exhibited in the user interface are very similar to those of KDD introduced in the previous section. MUSKUP consists of three main parts. These are database connection, data preparation and model development. Each button in these parts represents a task and takes the user either to a form or to a wizard related to this task. As it is important to do the tasks in order of precedence, the interface was designed to indicate this order. Forms accessed through the user interface make it possible to follow the KDD process and to perform the database management activities therein orderly and easily.

### 2.3. Architecture of MUSKUP

Fig. 3 shows the MUSKUP's working principle as a block diagram. The user can access to forms directly and indirectly for eight different tasks (table management, sampling, cleaning, transformation, research and exploring, splitting, modeling, and control and validating).

In MUSKUP, at first a database connection is made to the data set on which includes the data that an investigation is to be carried out. After the connection established, various editing and backup activities listed below can be performed through table management form or sampling form, which can be accessed via table management form.

Depending on the data being mined, table management techniques listed here:

- Select specific columns to include in a new table (i.e. model table, test table).
- Drop tables from the data set that are no longer useful. It is used in cleaning.
- Create a copy of an existing table but include fewer rows. This is used to reduce the test time for find best modeling.
- Create a copy of an existing table but force the sampling algorithm to include a higher percentage of positive predicted values than actually existed in the original table. It is used in splitting step.

Next steps are cleaning, transformation, and data analysing (research and exploring) tasks that constitutes the data preparation part. In the cleaning process, following three tasks are performed.

- For the columns containing a large number of null values; in order to decide which columns to include or exclude in the modeling, the percentage of null values for each column is calculated. If the column has more than 80% null value, that column is excluded.
- For the columns that has too few or too many distinct values; mean, minimum and maximum value and the number of distinct values for each column are calculated. This information is used to exclude the columns which seem to be useless.
- For the values that are not compatible with normal distribution; the outlying values are determined and the rows containing them are marked. Then according to the solution adopted, the rows can be excluded from analysis or the related values can be replaced by the average value of those columns.

By using the *Data Transformation Services* (*DTS*) *import/export* assistant in SQL Server, two types of transformations were carried out. Firstly, if the input columns have too many distinct values (categories), it is difficult to find out how this affects the target column. The way to overcome this problem is to reduce the number of distinct values in the input columns. Thus, the input columns having a large

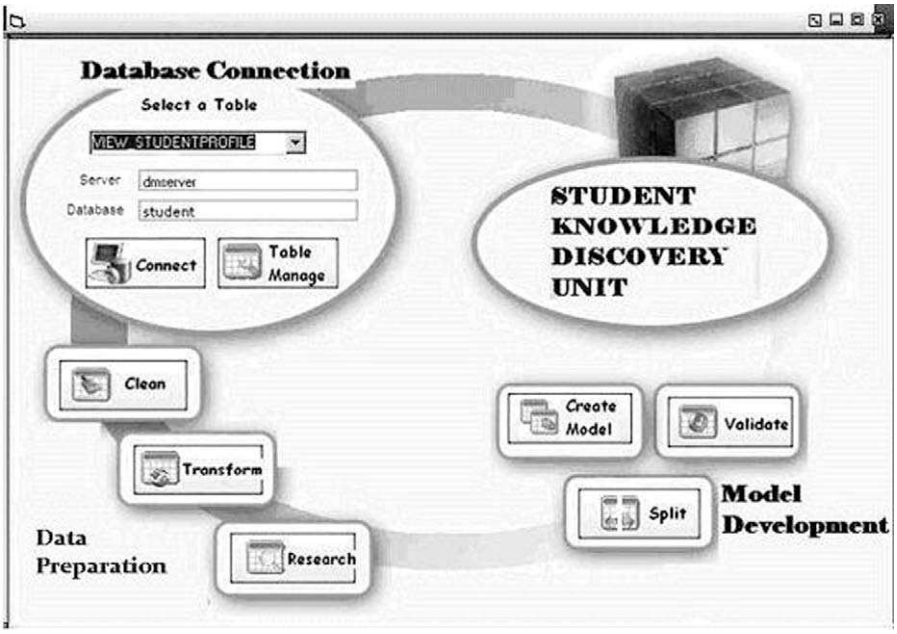



**Fig. 2.** MUSKUP user interface.

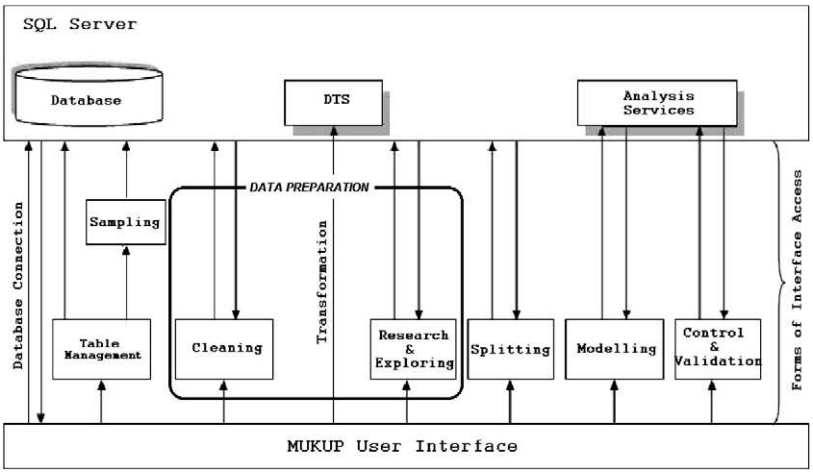



**Fig. 3.** MUSKUP architecture.

**Table 1**
The columns transformed and the corresponding new columns.

| Transformed columns | | | New columns after transformation | | |
|---|---|---|---|---|---|
| Column name | Data type | Number of distinct values | Column name | Data type | Number of distinct values |
| CITY | Varchar | 88 | CITY2 | Varchar | 8 |
| LIVECITY | Varchar | 325 | LIVECITY2 | Varchar | 8 |
| PREFERNO | Smallint | 24 | PREFERNO2 | Smallint | 6 |
| FMINCOME | Float | 249 | FMINCOME2 | Float | 5 |
| BIRTHDATE | Datetime | 2699 | AGE[a] | Decimal | 31 |
| REGDATE | Datetime | 135 | | | |
| GPA | Float | 349 | TARGET_A | Decimal | 2 |
| | | | TARGET_B | Decimal | 2 |

[a] Type 2 transformation (AGE = REGDATE − BIRTHDATE).

number of possibilities for the values are categorized exhaustively into a limited number of categories such as category 1, category 2, . . . , category *k*. This type of transformation is called Type 1. Secondly, in the cases that the variables which may have some effect on target column are not in a single column in the available columns, transformations on input columns can be used to create new useful columns. Also we used such a transformation, called Type 2. Table 1, shows all the transformations performed in this study.

In the model development part; the data set split into training and test data sets. A model is then formed by using the training data set. Finally, the validity of the model is checked by using test data set during the validation process. To accomplish these processes, SQL Server is mainly utilized in data base management activities, and Analysis Services is primarily utilized in modeling and validating. Section 2.4 includes some extra information about these steps.

Splitting data allows user to create a model and to test the model by using data coming from the same source. Two new tables are created as training and test table. Then 70% and 30% of main table's rows are divided randomly into the two new tables, training and test tables, respectively.

MUSKUP calculates the percentages of positive and negative values in the target column (e.g. in Model I: TARGET_A, GPA below 2.0 is negative, above 2.0 positive) and displays these results on a data splitting form. The resulting percentages enable us to see to what extent new tables represent the data in the main table.

### *2.4. Classification of students by using decision trees*

Classification is the process of assigning new objects to one of the pre-defined the classes after examining their characteristics (Michalski & Stepp, 1983). In the student classification process, each student will be allocated to a pre-defined group (successful/unsuccessful) according to his/her own success characteristics. In the classification by means of decision tree, classifications models are established through database development and analysis of the data is obtained by induction.

The models obtained help us to understand the existing data and to predict the behaviors of new incoming data. Producing rules in the tree structure provides simple and easy-to-understand models (Berry & Linoff, 2000). Operations can be carried out even with a little information preparation (Quinlan, 1986). Therefore, decision trees can easily be integrated with information technologies and renders high level of automation possible.

In this study, the classification of students was made according to their own success characteristics and decision trees were preferred for the model as they help produce more understandable results. Therefore, Microsoft Decision Trees (MDT) algorithm in Microsoft SQL Server Analysis Services was applied to create decision tree models. Soni and his friends (2002) conducted a study that showed the performance of MDT. Lift chart method was preferred in the evaluation of the efficiency of the models.

**Table 2**
Correlation matrix.

| Target columns | Input columns | | | |
|---|---|---|---|---|
| | KEY | HIGHGRADE | HIGHGRAD1 | PREFERNO |
| TARGET_A | −0.0268 | 0.0438 | 0.0022 | 0.0462 |
| TARGET_B | −0.0309 | −0.0274 | 0.0134 | 0.0466 |
| | PREFERNO2 | FMINCOME | FMINCOME2 | IDCODE |
| TARGET_A | 0.0451 | −0.0173 | 0.0066 | 0.045 |
| TARGET_B | 0.049 | 0.0468 | 0.0487 | −0.02 |
| | PROGTYPE | SEMCOUNT | SEMRECNO | YEARECNO |
| TARGET_A | 0.0073 | 0.0178 | −0.0025 | 0.2069 |
| TARGET_B | 0.0221 | −0.0135 | 0.0184 | −0.0073 |
| | TARGET_A | TARGET_B | | |
| TARGET_A | 1 | 0.308 | | |
| TARGET_B | 0.308 | 1 | | |

**Table 3**
The columns and their definitions included to Models 1 and 2.

| No. | Column name | Data included | Type of data | No. | Column name | Data included | Type of data |
|---|---|---|---|---|---|---|---|
| 1 | KEY | Key column (1, 2, 3, ...) | Single | 13 | LANGPREP | Foreign language to be learned in prep-school | Varchar |
| 2 | TARGET_A[a] (target for Model I) | If the GPA is 2.0 or more 1, if not 0 | Single | 14 | PROGTYPE[b] | Type of the university program | Integer |
| 3 | TARGET_B[b] (target for Model II) | If the GPA is 3.0 or more 1, if not 0 | Single | 15 | HIGHGRAD1[b] | Finishing the high school with the best average | Integer |
| 4 | REQPREP | Request for preparatory school (Y/N) | Varchar | 16 | SEMRECNO[b] | The number of the terms spent at school | Integer |
| 5 | MILITARY | State of military service (do or not) | Varchar | 17 | SEMCOUNT | Total of terms | Integer |
| 6 | GENDER | Gender (M/F) | Varchar | 18 | YEARECNO[a] | The number of the years spent at school | Integer |
| 7 | GRANTPTF | Grants to pay tuition fee (receiving/not receiving) | Varchar | 19 | FMINCOME | Monthly income of the family | Single |
| 8 | EDUCTYPE | Type of the education (day time/evening classes) | Varchar | 20 | CITY2 | The region in which the city where the student was born is located | Varchar |
| 9 | DEPTNAME | Department name | Varchar | 21 | LIVECITY2 | The region in which the city where the student lives is located | Varchar |
| 10 | IDCODE | Department and type of education together | Varchar | 22 | PREFERNO | The place of the university attended among the preferences done in the university entrance exam | Integer |
| 11 | REGTYPE | Type of registration to the school | Varchar | 23 | PREFERNO2 | The place of the university attended among the preferences done in the university entrance exam | Integer |
| 12 | TYPEHIGH | Type of the high school | Varchar | 24 | FMINCOME2[b] | Monthly income of the family (four categories grouped) | Single |

[a] Included to the 1st model only.
[b] Included to the 2nd model only.

### *2.5. Software technologies used in knowledge discovery*

Window Server 2003 and SQL Server 2000 were used as an operating system and a DBMS, respectively. In addition, Analysis Services in SQL Server were used to create and validate DM models, MDAC 2.8 was used to have access to data, Angoss OLE DB (Data Mining Consumer Controls) was used to display the models and validation results.

Furthermore, MUSKUP software was developed by using the programming technologies of Microsoft ActiveX Data Objects (ADO) and Decision Support Objects (DSO) both of which came with Visual Basic 6.0. FlexGrid control was used in order to examine the data in tables, and chart controls were used to explore the data graphically. These controls are also included in Visual Basic 6.0 toolbox.

While implementing some tasks related to KDD process, SQL server tools were called on. For example, in the transformation step, "DTS import/export wizard" accesses the data set and then transforms the column values. Moreover, DTS creates predictions based on DM model and performs actions according to the results.

## 3. Results of knowledge discovery in databases

Results for every stages of discovery process performed on the students data are presented below.

### *3.1. Correlations of some of the input columns with target columns*

While developing the models, columns will effect the result would be preferred. Correlation matrixes can be used to find such columns having only numeric values.

Table 2 gives correlations of target columns with some of the input columns. These correlations indicate degrees of relationship between target columns and the potential predictor input columns.

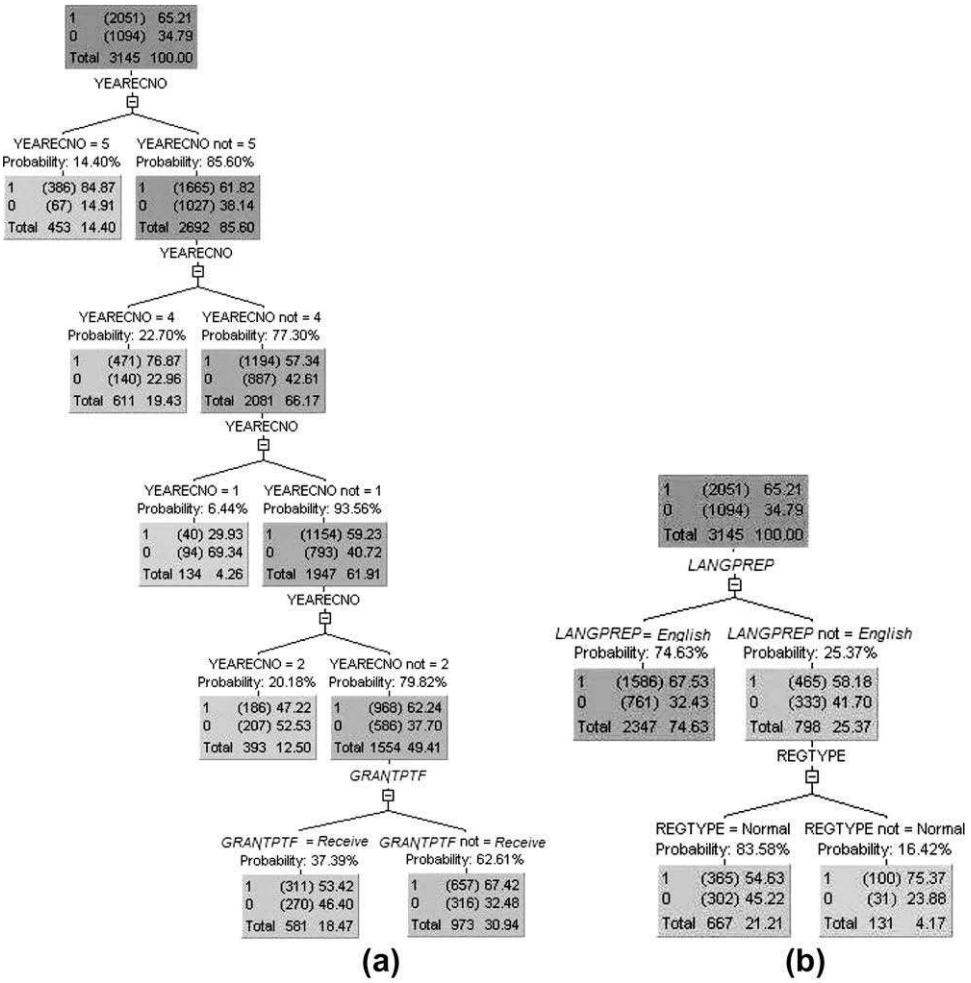



**Fig. 4.** Graphical display of the decision trees for Model I.

The reason for not giving correlations in all fields of the database used in Table 2 is the presence of some columns in the student database containing null, missing, inconsistent data and also some columns such as mother name, type of blood, which cannot be associated logically with the target variable. Elimination of these fields is a part of cleaning process, which is done in order to run MDT algorithm more efficiently.

So, the values of ±0.01 was accepted to be the lowest limit in the correlation matrix, That the columns whose correlations with target columns are below this value were ignored. Thus, the columns, namely; HIGHGRAD1, FMINCOME2, SEMRECNO and PROGTYPE were not included in Model I and the column YEARECNO was not included in Model II according to the criterion. In addition to this, since they are used in the model development, the key column (KEY) and the target columns (TARGET_A and TARGET_B) themselves were exempted from this rule.

### *3.2. Columns used in modeling and decision tree views*

Table 3 shows the columns and their definitions included to the models.

The columns affecting the target column are given in the first decision tree in Fig. 4a, in the order of importance: the column YEARECNO and the column GRANTPTF.

As seen from Fig. 4a, separations happened in the column YEARECNO values: 5, 4, 1 and 2, respectively. The prediction is better for the values 5 and 4 of YEARECNO. This is not surprising because students are to complete their education in between 4 and 7 years with a requirement of GPA $\geqslant$2.00. As another important item, GRANTPTF slightly affected the target and caused two points increase in probability for prediction purposes. One thing to notice is that those students who do not receive grants to pay their tuition fees are more likely to be in the positive value group, namely 1 indicating GPA $\geqslant$2.00.

When the impact of YEARECNO is eliminated from the model, the resulting decision tree appears in Fig. 4b. Here, the most affective columns used to predict the target column are LANGPREP and REGTYPE. Since LANGPREP is English and REGTYPE is different from the normal type such as transferred from another university, these are seen to be influential on student success in decision tree. Students transferred from another university must go through the adaptation training for a year and must be successful to some certain extent, and these are also influential on32;uation, if all of the columns holding null values were not deleted. Fig. 5b shows the resulting decision tree when these columns are eliminated from the model. The decision tree algorithm selected FMINCOME2 as the most important factor that determines the split on the data. The column FMINCOME2 contains categorized data on monthly incomes of the students' family with the value 3 corresponding to the middle level income (400–600 Turkish Liras) group. It is thus concluded that high level of success is more correlated with the middle level income.

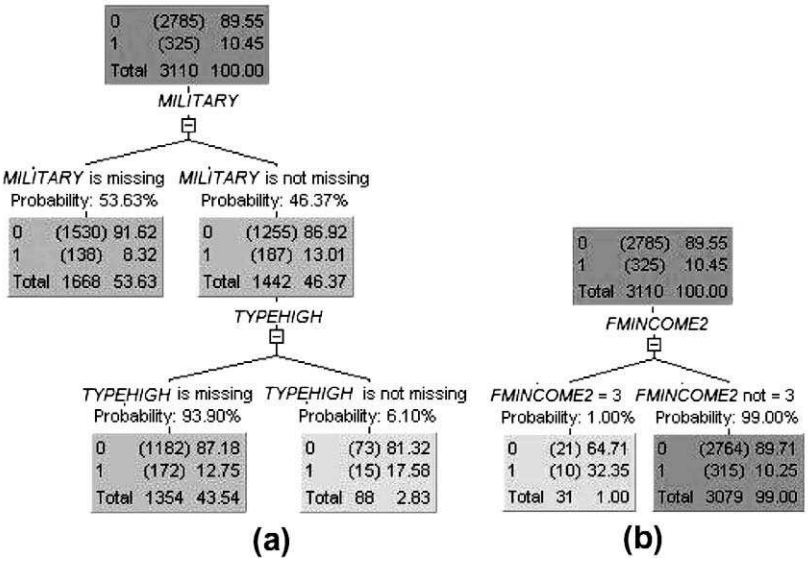



**Fig. 5.** Graphical display of the decision trees for Model II.

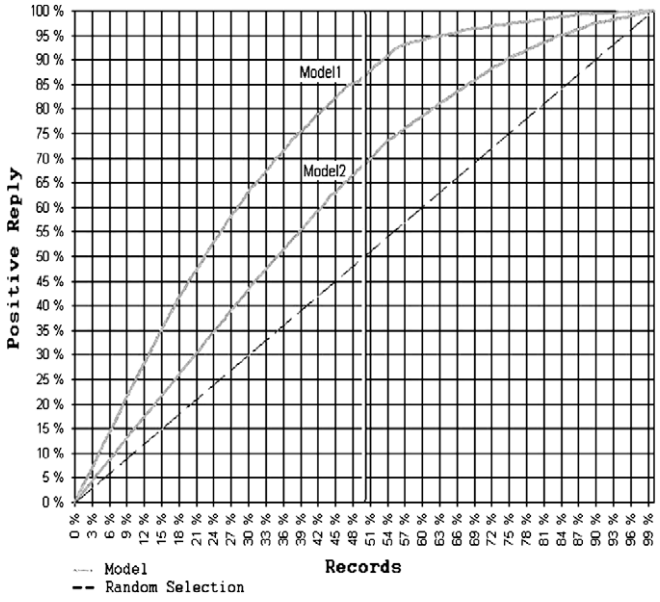



**Fig. 6.** Lift graphics for the models.

### 3.3. Model validation and lift graphics

Lift graphics for the models are given at Fig. 6.

When 50% of all existing records are selected randomly, 50% of all the positive responses can be reached. This is a characteristic curve since the number of total positive responses in the records is always fixed. When trained models are employed, this value increases to 87% for Model I and 68% for the Model II. Accordingly the lift value becomes 87/50 = 1.74 for Model I and 68/50 = 1.36 for Model II. This indicates that these models have prediction capacity. The reason for lower prediction potential of Model II is that Model II has less positive values in the target column compared to Model I. For Model I, the positive value means that GPA is greater or equal to 2.0, whereas the positive value for Model II means that GPA is greater or equal to 3.0. Model I has a 65% positive value ratio. For Model II, this ratio is 11%. Thus small positive ratio makes it difficult to establish relationships during training of the model.

## 4. Conclusions and suggestions

The paper presents a knowledge discovery applied on university students' demographical data. In order to explore the factors having impact on the success of university students, knowledge discovery software, called MUSKUP, has been developed and tested on this data. With this system, all the tasks involved in the knowledge discovery process are performed collectively.

In evaluating student performance, decision tree classification technique has been carried out. The classifications attempt to find out which demographic data is most influential on student GPA. The scope of study is limited to determining profiles of students whose

GPA is equal to 2.0 (which is the minimum GPA required for graduation) or greater and of those students whose GPA is equal to 3.0 or greater (honor degree). Thus, two classification models were obtained. In the first model, the types of registration to the university and in the second model, the monthly income of the family were found to be the greatest factors affecting the target. In checking performances of the models, lift graphics were used. According to the lift graphics, values 1.74 for Model I and 1.36 for Model II were found, respectively. This shows that the models have prediction capability. Missing data of some columns in dataset influenced the success of the system directly. Therefore, if the amount of data and the number of variables is increased, more accurate predictions about student success can be made.

The parts of the data prepared and the model developed for MUSKUP were designed considering the needs of the used data and the problem investigated. Therefore, it will require modifications and extensions for different data and problems. Whereas, this study may help other studies to be carried out in the future where KDD processes will be integrated with DBMSs of organizations.

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# 1. FIRST STEP: PROJECT TITLE SELECTION

**Project Title:** Student Performance Analysis System

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The project focuses on maintaining student, subject and assessment-mark records and using the stored marks to produce clear academic performance summaries.

# 2. PROJECT TITLE BASE PAPER

Base Paper: ?A new student performance analysing system using knowledge discovery in higher educational databases?
Authors: Huseyin Guruler, Ayhan Istanbullu and Mehmet Karahasan
Journal: Computers & Education, Volume 55 (2010), pages 247?254.
The complete base paper is included after the project report.

# 3. PROJECT ABSTRACT

## 3.1 Introduction

Student Performance Analysis System is a browser-based mark-register application intended to help organize student records, subjects, assessment marks and performance reports. It provides a dashboard for class-level indicators and a report section for identifying top-performing students and students whose results fall below a chosen percentage threshold.

## 3.2 Abstract

The system allows users to add, view, edit and delete student records, subjects and marks. It calculates each student’s percentage and grade, checks whether the student meets the pass mark in every subject, and displays subject averages, pass percentages, mark distributions, topper lists and low-performer lists. The supplied implementation is built with HTML, CSS and JavaScript and stores its records in the browser’s local storage. This makes it suitable as a small demonstration or prototype; a shared institutional deployment would require a server-side backend, a central database, authentication and appropriate access controls.

## 3.3 Existing System

In a traditional mark-register workflow, marks may be maintained in paper registers or separate spreadsheets. Such approaches can work for small classes but may require repeated manual calculations and manual preparation of summaries. The uploaded website provides a more organized browser interface for recording marks and generating summaries. The report does not assume that every institution currently uses the traditional approach; it describes it as a common comparison model.

## Disadvantages of the Existing System

- Manual calculations can take time and may introduce arithmetic or transcription errors.
- Separate paper sheets or spreadsheets can make it harder to maintain consistent records.
- Preparing class averages, subject pass rates and topper lists may require repeated work.

- Identifying students below a performance threshold may depend on manually reviewing marks.
- Sharing and controlling access to records can be difficult when files are copied between users.

## 3.4 Proposed System

The proposed Student Performance Analysis System provides one interface with Dashboard, Students, Subjects, Marks and Report sections. Student records contain a roll number, name and section. Subject records include the subject name, maximum marks and pass marks. Mark records link a student with a subject and assessment, and the application checks that entered marks are within the subject's maximum.

- Create, view, update and delete student records.
- Create, view, update and delete subjects and their mark limits.
- Enter, edit and delete assessment marks for students.
- Calculate total marks, percentage, grade and pass/fail status.
- Display class average, students passing all subjects, overall pass rate, subject averages and subject pass percentages.
- Show a configurable topper list and a list of students below a selected percentage threshold.

## Advantages of the Proposed System

- Reduces repetitive manual calculation for the indicators supported by the application.
- Presents student, subject and mark information in consistent tables.
- Makes performance patterns easier to review through dashboard charts and reports.
- Allows records to be corrected through edit functions and removed when necessary.
- Supports basic identification of students who may benefit from additional academic support.

## 3.5 Extension

The following are recommended future extensions, not features currently present in the uploaded HTML implementation:

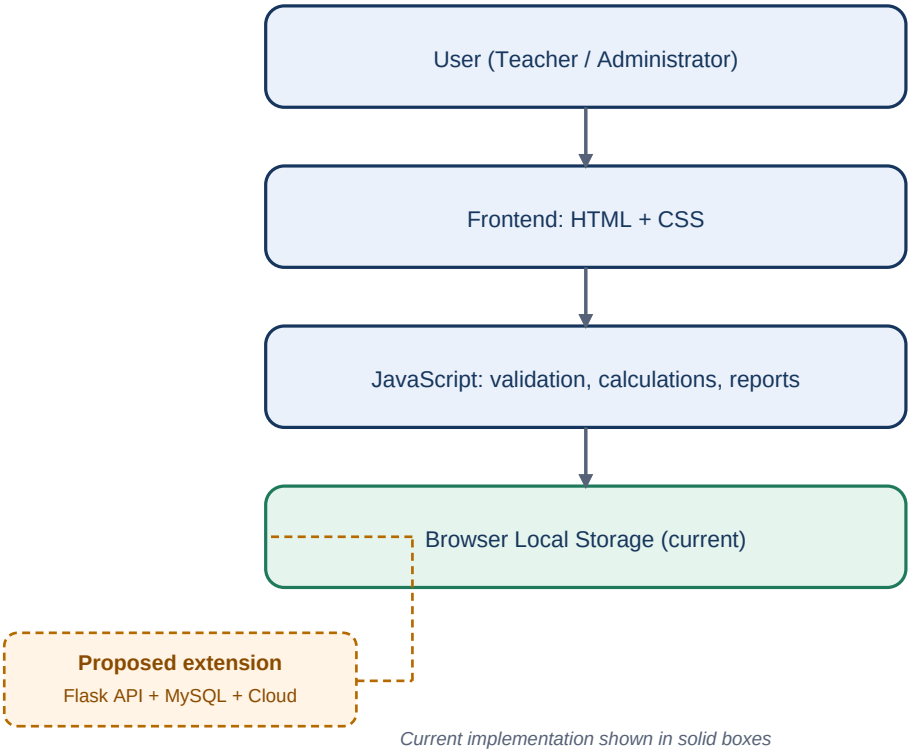
- Add a Python Flask backend and a MySQL database so records can be centrally stored and accessed by authorized users.
- Introduce login and role-based permissions for administrators, teachers and students.
- Add attendance, assignment completion and term-wise progress tracking if the institution needs these measures.
- Provide exportable PDF/CSV reports, date filters and comparisons across terms.
- Deploy the application to a secure cloud hosting service with backups and monitoring.
- Add optional trend analysis to flag changes in performance over time, with teachers reviewing results before making decisions.

## Advantages of the Extension

- Centralized data can be shared across authorized devices instead of being limited to one browser profile.
- Authentication and permissions can improve confidentiality and accountability.
- Backups and cloud deployment can improve availability when correctly configured.
- Term-wise trends and exports can support more informed academic review.

## 3.6 System Architecture

The diagram below separates the current browser-based prototype from a recommended full-stack extension. In the current version, the browser interface and JavaScript logic operate on data stored in local storage; a Flask/MySQL/cloud layer is a proposed enhancement.




*Current implementation shown in solid boxes*

## 3.7 Requirements

### Hardware Requirements

- Computer or laptop capable of running a modern web browser.
- At least 2 GB RAM recommended for a basic prototype; 4 GB or more is preferable for comfortable development.
- Keyboard, display and sufficient storage for the browser and project files.
- Internet connection is not required for the core local HTML prototype after the file is available; it is required for online hosting or remote access.

### Software Requirements

- Operating system: macOS, Windows or Linux.
- Web browser: a recent version of Chrome, Firefox, Safari or Edge.
- Frontend: HTML5, CSS3 and JavaScript.
- Current data storage: browser localStorage.
- Development tools: a text editor or IDE such as VS Code.
- For the proposed extension: Python 3, Flask, MySQL Server and a database connector; a cloud hosting account and HTTPS configuration for deployment.

# 4. CONCLUSION

The uploaded Mark Register website demonstrates the main workflow of a Student Performance Analysis System: maintaining student and subject information, recording assessment marks, calculating percentages and grades, and presenting performance summaries. Its dashboard and reports can help users review subject-level results, rank students and identify results below a chosen threshold. Because the current version stores information in the browser, it is best considered a prototype rather than a multi-user institutional system. A backend, central database, authentication and secure deployment would be needed for broader use.

# 5. FUTURE SCOPE

- Convert the prototype into a full-stack application using Flask and MySQL.
- Provide secure sign-in and separate permissions for teachers, administrators and students.
- Add attendance, assignments and term-wise progress where required.
- Generate downloadable PDF and spreadsheet reports.
- Deploy through a cloud service with backups, monitoring and HTTPS.
- Add trend charts and carefully reviewed early-support indicators based on approved academic data.
- Test usability, input validation, privacy and correctness with institution-approved sample data before production use.

## REFERENCE NOTE

Primary project reference: user-uploaded file “mark-register.html” (Mark Register website). The HTML source includes Dashboard, Students, Subjects, Marks and Report sections; localStorage persistence; grade and pass/fail calculations; subject summaries; topper ranking; and a low-performer threshold report. A separate online base paper was not supplied, so formal bibliographic details should be added after selecting an approved publication.